

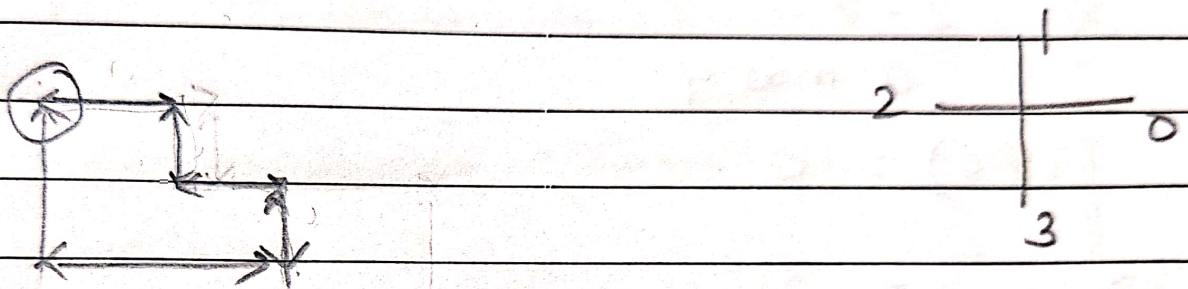
4 direction chain code $\Rightarrow \text{mod } 4$

8 direction chain code $\Rightarrow \text{mod } 8$

For chain code $c(i)$ of length n

$$CFD(i) = (c(i+1) - c(i)) \text{ mod } N$$

where $N = 4$ (for 4 direction)



Index = 1 2 3 4 5 6 7 8

chain code = 0 3 0 3 2 2 1 1

$$(1+1)=2 \quad c(1)=0$$

$$\textcircled{1} \quad c(1) = (c(i+1) - c(i)) \text{ mod } 4$$
$$(3 - 0)$$

$$3 = 3 \text{ mod } 4$$

$$c(1) = 3$$

$$\textcircled{2} \quad c(2) = (c(3) - c(2)) \text{ mod } 4$$
$$= (0 - 3)$$

$$1 = -3 \text{ mod } 4$$

$$c(2) = 1$$

$$\begin{aligned} (3) \quad &= 3 - 0 \\ &= 3 \bmod 4 \end{aligned}$$

$$c(3) = 3$$

$$\begin{aligned} (4) \quad &2 - 3 \\ &= -1 \bmod 4 \end{aligned}$$

$$c(4) = 3$$

$$\begin{aligned} (5) \quad &2 - 2 \\ &= 0 \bmod 4 \end{aligned}$$

$$c(5) = 0$$

$$\begin{aligned} (6) \quad &1 - 2 \\ &= -1 \bmod 4 \end{aligned}$$

$$c(6) = 3$$

$$\begin{aligned} (7) \quad &1 - 1 \\ &= 0 \bmod 4 \end{aligned}$$

$$c(7) = 0$$

(8) last element is circular.

$$CFD(n) = (c(1) - c(n)) \bmod 4$$

$$= 0 - 1 \bmod 4$$

$$c(8) = 3$$

$CFD = 31330303$

CFD is rotation invariant

★ Rotation adds a constant direction offset to all chain codes.

★ Differences betⁿ consecutive directions remove that constant.

so, CFD is rotation invariant.

Still dependent on starting point

So, Normalized CFD (NCFD)
all circular shifts of 31330303

13303033

33030331

30303313

03033133 ✓

30331330

03313303

33133030

Smallest

CED removes rotation,
Normalization removes starting point
together they give a unique
Shape signature.

shortcut - start NCFD from smallest
digit & continue circularly.